GPU COMPILER · v0.1.0

Deuteron

Scheduled Emmy IR → NVIDIA PTX

Guarded compiler passes for RTX 5090.

Tensor cores

BF16x3 + TF32x3
FP32 accumulation

Async copies

Global → shared memory
Commit / wait groups

FP16 storage

Half storage
Explicit FP32 arithmetic

From schedule to GPU code

Program structure and rewrite guards drive the transformations.

01
Scheduled IR
Tiles, loops, memory
+ launch metadata

→

02
Deuteron passes
Validate and transform
Check rewrite legality

→

03
LLVM + NVPTX
Optimize typed LLVM IR
Emit NVIDIA PTX

→

04
CUDA → GPU
Assemble and launch
RTX 5090 / SM120

A few passes worth highlighting

Vectorize loads
Coalesced memory access

Reuse registers
Avoid repeated global loads

Reduce trees
Specialize warp / CTA sums

Tensorize
Lower eligible contractions

Tensor cores. Copies. Types.

TENSOR CORES

BF16x3
TF32x3

FP32 inputs + accumulation

`mma.sync` / `ldmatrix`

Compensated three-product approximations

Also: ordinary FP32 products

ASYNC COPIES

4 / 8 / 16

bytes per transfer

Global → Shared

`cp.async`

Commit / wait groups
CTA / warp barriers

FP16 STORAGE

Store FP16.
Compute FP32.

FP16 → FP32 → FP16

Exact widening
Nearest-even narrowing

BF16 / TF32 are internal tensor formats. FP16 uses the documented pointwise subset.

Inspect the generated PTX

A real TF32x3 matrix compile exposes both instruction families.

COMPILE FOR SM120 · CPU ONLY

```
python compile.py examples/tensor_matmul/matmul_tiled_128.ir.json \
  --arch sm_120 --opt-level 3 --contraction-precision tf32x3 > /tmp/demo.ptx
```

TENSOR-CORE OPCODE

```
mma.sync.aligned.m16n8k8.row.col.f32.tf32.tf32.f32
```

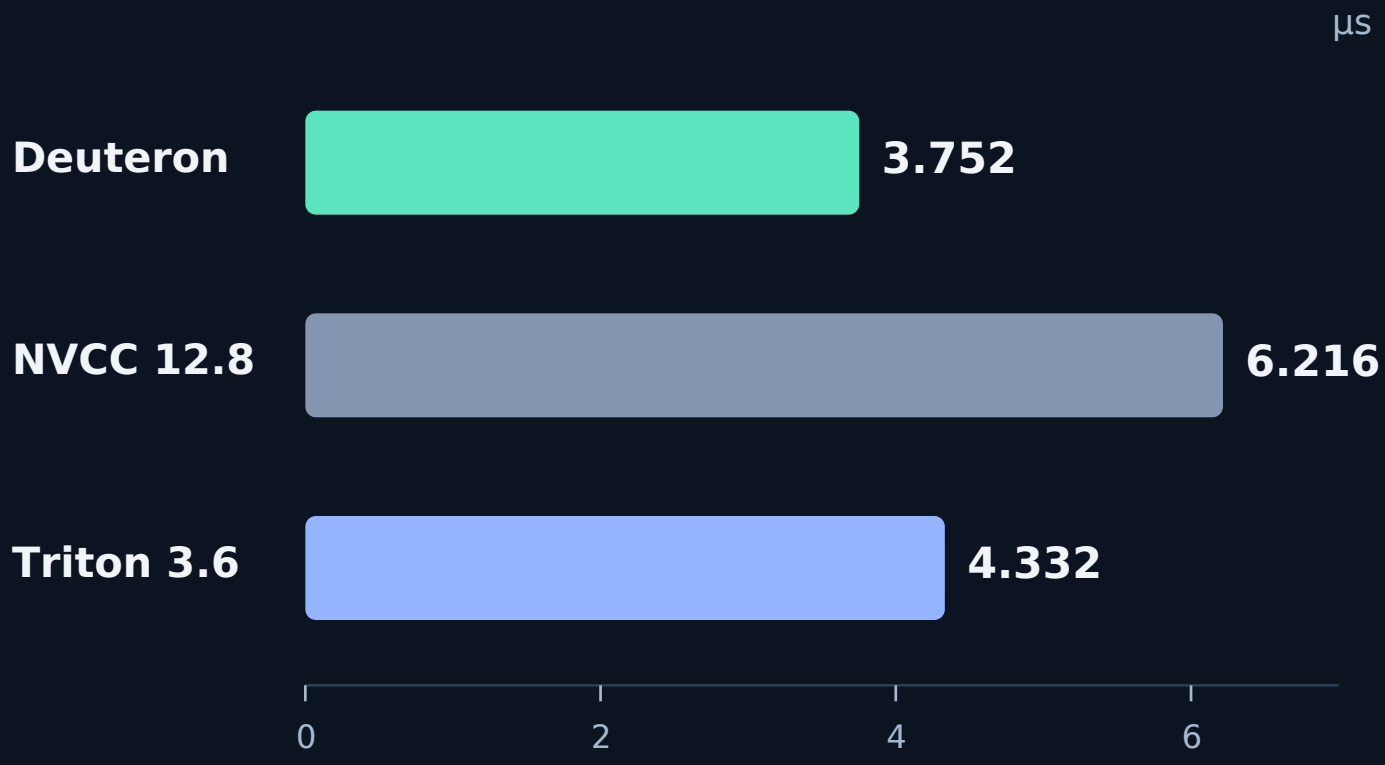
GLOBAL-TO-SHARED COPY

```
cp.async.cg.shared.global [%r1], [%rd1], 16;
```

Actual emitted PTX. MMA operands omitted for legibility. This demonstrates compilation.

Measured on RTX 5090

Geometric mean warm latency · 15 open workloads · lower is better



1.66×

vs generated CUDA / NVCC

1.15×

vs tested Triton baseline

270 correct warm calls · 20 warmups / 200 iterations · Original inputs and tolerances

Selected baselines; compilation and transport excluded. Full protocol and raw results are included.

A compiler you can inspect

Repeatable builds, explicit contracts, and recorded evidence.

103/103

H100 regression tests

11/11

RTX 5090 feature examples

3.12 + 3.13

Verified Python installs

Restricted parsing + ABI validation

Bounded caches + atomic records

Installable CLI + concurrent compilation

Documented scheduled-IR subset

Explore the examples. Inspect the PTX. Reproduce the results.